

Do Elicited Epistemic States Predict Error?

A Preregistered Predictive-Validity Test of Three-Component Verbalized Uncertainty in Large Language Models

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Abstract

Several formalisms propose representing a language model’s epistemic state with more than one number: degrees of truth, indeterminacy and falsity as separate components that need not sum to one. Prior work has shown that models readily emit such triples and that the components frequently exceed unit mass. What has not been established is whether the elicited triple carries operationally useful information. We test this directly and under a preregistered decision rule: holding the prediction target fixed at answer correctness, does an elicited (T, I, F) triple predict error better than simpler signals obtained from the same model? We compare five signal families — the elicited triple, verbalized scalar confidence, self-verification $P(\text{True})$, sequence log-probability, and discrete semantic entropy — on four models from four vendors and four datasets spanning free-form generation (TriviaQA, NQ-open), closed-label classification with clean labels (SciQ as four-option multiple choice) and closed-label classification with genuinely ambiguous items (ChaosNLI, 100 human labels per item). Every elicited arm is run under three elicitation protocols and reported as a range. The result is negative. Across 48 model $\times$ dataset $\times$ protocol cells, the best-case triple (a logistic combination fitted on a held-out calibration split) does not outperform a single verbalized confidence number (mean paired AUROC difference -0.010 ; the 95% interval excludes zero in favour of the triple in 2 cells and against it in 6), and the indeterminacy component adds no information over truth and falsity alone (mean difference $+0.003$). Semantic entropy outperforms every elicited signal on free-form generation ($+0.11$ to $+0.21$ AUROC), and at a guaranteed selective risk of 0.10 the triple yields lower achieved coverage than the scalar in 23 of 36 cells. A post-hoc analysis on ChaosNLI shows that neither the elicited nor a sample-constructed indeterminacy tracks human label disagreement better than a scalar. We conclude that elicited three-component epistemic states do not carry predictive information beyond a single elicited number, and that arguments built on the triple as a measurement instrument lack predictive support. Prompts, raw generations, gradings, the sealed analysis plan and analysis code accompany this submission.

1 Introduction

A language model asked a question it cannot answer has, in principle, several distinguishable predicaments. It may lack the information. The question may be vague. The question may be about a future contingency. The proposition may be paradoxical. A single scalar confidence score compresses all of these into one number, and a normalized output distribution forces whatever mass is withheld from one answer onto the others.

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This observation has motivated a family of proposals that represent the model’s state with a triple rather than a scalar: degrees of truth, indeterminacy and falsity treated as separate and not required to sum to one [Leyva-Vázquez and Smarandache, 2026]. Empirically it is established that models will emit such triples when asked, that the components frequently sum to more than one, and that the phenomenon replicates across vendors [Mason, 2026]. It is also established, by the same replication, that the scalar triple is *insufficient* to separate paradox from ignorance from contingency: distinct predicaments can produce identical triples.

None of this answers the question a practitioner would ask. Suppose we accept that the triple is a richer description. Does it *do* anything? Concretely: if we want to know which of a model’s answers will turn out to be wrong, does the triple tell us more than simply asking the model how confident it is?

That is the question this paper tests. We fix the target — $Y = 1$ if the model’s answer is incorrect — and require every candidate signal to predict the same Y on the same items with the same evaluation protocol. This turns a representational claim into a measurable one.

Why the comparison must include verbalized scalar confidence. The decisive baseline is not semantic entropy. It is asking the model for one number. If a three-component elicitation does not beat “how confident are you, 0–100?”, then the additional components are contributing vocabulary rather than information, and the representational argument has no empirical support at the level of prediction. To our knowledge this comparison had not been run. We treat it as the primary contrast and every other signal as a reference point.

Why protocol variation is part of the design, not a robustness check. Verbalized confidence is strongly sensitive to elicitation format [Tian et al., 2023, Xiong et al., 2024, Kim and Kang, 2026]. Any comparison between an elicited triple and an elicited scalar that uses one prompt per arm is confounded by prompt choice. We therefore run every elicited arm under three protocols that vary wording, scale and response format, and report the full range.

Contributions.

1. A preregistered predictive-validity test of elicited three-component epistemic states against four baselines under a single fixed target and evaluation protocol, with the analysis plan and all deviations sealed in a version-controlled repository before collection (Sections 4–6).
2. An explicit decisive contrast against verbalized scalar confidence, and an ablation isolating the contribution of the indeterminacy channel I over T and F alone.
3. Coverage of three task regimes — free-form generation, closed-label classification with clean labels, and closed-label classification with human-measured ambiguity — so that the indeterminacy component is tested against a target it could plausibly track.
4. A decision-relevant evaluation: conformal abstention at a fixed target risk, reporting achieved coverage per signal, rather than discrimination metrics alone.
5. A negative result, stated as such: the triple does not beat a single number, and I adds nothing, in every regime tested.

What we are not claiming. We do not test whether three-component representations are philosophically preferable, nor whether they describe internal states [Mason and Anand, 2026]. We test one thing: whether the elicited numbers predict error. The negative result does not refute the representational argument; it establishes that the argument cannot currently be supported on predictive grounds.

2 Background and related work

Uncertainty quantification for LLMs. The dominant sampling-based signal is semantic entropy, which samples multiple generations and measures dispersion over meaning-equivalence classes rather than surface strings [Kuhn et al., 2023, Farquhar et al., 2024]. Self-verification approaches ask the model to judge its own candidate answer, yielding $P(\text{True})$ [Kadavath et al., 2022]. Sequence-level token probability remains a strong and cheap baseline where log-probabilities are exposed. Conformal prediction supplies distribution-free, finite-sample guarantees and is now routinely combined with these scores to produce accept/abstain rules [Angelopoulos and Bates, 2021, Karimi et al., 2026]; selective prediction and the risk–coverage curve give the decision-relevant framing [Geifman and El-Yaniv, 2017].

Verbalized confidence and its fragility. A parallel line asks the model to state its confidence in words or numbers. Verbalized confidence from RLHF-tuned models can be better calibrated than token probabilities in some settings [Tian et al., 2023], but it concentrates in a narrow high band regardless of correctness, and the reported number is governed by response scale and prompt structure [Xiong et al., 2024, Kim and Kang, 2026]. Probes on hidden states discriminate correctness better than the verbal channel on the same items [Ahdritz et al., 2024]. This literature is the reason our design treats elicitation protocol as a factor.

Decomposed uncertainty. A growing literature asks whether a model’s uncertainty can be split into components — epistemic versus aleatoric [Ahdritz et al., 2024], belief divergence versus belief conflict [Shi et al., 2026] — and whether models decline to answer when they should [Meng et al., 2026]. The three-component triple tested here is one instantiation of this idea; the question of whether such a decomposition can be *elicited* verbally, and whether it carries information beyond a scalar, is the gap this paper addresses.

Multi-component uncertainty formalisms. Representing belief with more than one number has several lineages: Dempster–Shafer belief functions [Shafer, 1976], subjective logic [Jøsang, 2016], imprecise probability, and four-valued Belnap–Dunn logics, each with an established treatment of ignorance and conflict. Neutrosophic (T, I, F) belongs to this family and is the specific instantiation tested here [Leyva-Vázquez and Smarandache, 2026]. We take no position on the relations among these formalisms; we evaluate one of them empirically, and the protocol applies unchanged to the others.

Positioning. The closest prior work established that unconstrained elicitation produces $T + I + F > 1$ at high rates and that this replicates across vendors [Leyva-Vázquez and Smarandache, 2026, Mason, 2026]. Neither establishes predictive validity.

3 What would count as evidence

Let x be a question, $\hat{a}(x)$ the model’s answer under greedy decoding, and $Y(x) = \mathbb{1}[\hat{a}(x) \text{ is incorrect}]$ under a fixed grading function. A *signal* is any map $s : x \mapsto \mathbb{R}$ intended to be monotone in the probability that $Y = 1$. Every signal in this study predicts the same Y , on the same items, from the same model, and is evaluated with the same metrics on the same split.

For a triple (T, I, F) we define three scalarizations, fixed in advance:

$$s_{\text{TF}} = F - T, \tag{1}$$

$$s_{\text{TIF}} = F + \frac{1}{2}I - T, \tag{2}$$

$$s_{\text{TIF}}^{\text{lr}} = \text{logistic combination of } (T, I, F) \text{ fitted on the calibration split.} \tag{3}$$

Equation (1) ignores I entirely; (2) charges indeterminacy at a fixed rate; (3) lets the data choose the weights and is the most favourable treatment of the triple. The contrast $s_{\text{TIF}}^{\text{lr}}$ versus s_{TF} isolates whether I carries information at all. Fitting (3) uses only the calibration split; all reported metrics are computed on the held-out test split.

4 Method

4.1 Signals

- A. Elicited triple.** One additional call per item requesting T, I, F in $[0, 1]$ with no normalization constraint, parsed to a triple. Scalarized by (1)–(3).
- B. Verbalized scalar confidence.** One additional call requesting a single integer 0–100 (or a verbal grade under P3). *Primary baseline.*
- C. Self-verification $P(\text{True})$.** One additional call presenting the question and the model’s own answer and asking whether it is true (binary under P1–P2; a stated probability under P3).
- D. Sequence log-probability.** Mean token log-probability of the greedy answer where the provider exposes it: available for all items of gpt-4o-mini, 30% of items of Llama-3.1-8B (provider-dependent), and none for Claude or Gemini.
- E. Discrete semantic entropy.** $K = 10$ samples at temperature 1.0, normalized and clustered into equivalence classes by answer normalization, then Shannon entropy over cluster frequencies. On closed-label datasets the classes are the labels themselves.

Arms A, B and C are elicited and therefore run under all three protocols. Arms D and E are sampling-based and have no elicitation protocol.

4.2 Elicitation protocols

Three protocols per elicited arm, varying wording, scale and response format simultaneously so that they are distinguishable formats rather than paraphrases: P1, terse numeric, no explanation; P2, explanation first, then numbers; P3, verbal grades (NONE . . . TOTAL) mapped to a fixed numeric grid, with the dimensions presented in reverse order relative to P1. Full texts in Appendix A.

4.3 Datasets

Four datasets, $N = 300$ items each, sampled with seed 20260823 and split 50/50 into calibration and test:

TriviaQA rc.nocontext validation [Joshi et al., 2017]; free-form, rich alias sets.

NQ-open validation split [Kwiatkowski et al., 2019, Lee et al., 2019]; free-form, harder.

SciQ-MC SciQ validation [Welbl et al., 2017] served as four-option multiple choice using the dataset’s own distractors, options shuffled with the fixed seed, the model answering with a letter. This is the closed-label regime with clean labels.

ChaosNLI SNLI and MNLI-matched items with 100 human labels each [Nie et al., 2020]; three labels; gold = majority label. Half of the sample is drawn from the lowest-entropy third of the pool (clear items) and half from the highest-entropy third (genuinely ambiguous items), so that the indeterminacy channel can be checked against measured human disagreement. This is the closed-label regime with ambiguous labels.

Short-form and closed-label tasks are a deliberate restriction: they make Y well defined without an LLM grader.

4.4 Models

Four models from four vendors, accessed through OpenRouter on 1 September 2026 with temperature 0 for greedy and elicited calls: `openai/gpt-4o-mini`, `anthropic/claude-haiku-4.5`, `google/gemini-2.5-flash`, `meta-llama/llama-3.1-8b-instruct`.

4.5 Grading

Y is computed by normalized exact match against the gold answer set (lower-case, strip articles and punctuation, alias expansion; letter or option text for SciQ-MC; label word for ChaosNLI). Items where the model refuses to answer (I DON’T KNOW) are recorded separately and excluded from the discrimination analysis: a signal cannot be credited for predicting the error of an answer that was never given. Refusal rates: TriviaQA 4.2%, NQ-open 16.3%, SciQ-MC 0.8%, ChaosNLI 0%; by model, 4.3–5.9%.

4.6 Cost

The realized design comprised 4,800 valid item-cells and approximately 106,000 API invocations, of which about 2,700 failed during a credit exhaustion in the first collection pass and were re-collected in full in a second pass (the pipeline is resumable and re-collects error rows; no partial records were used). Total spend was approximately US\$33.

5 Evaluation

Discrimination. AUROC and AUPRC for predicting $Y = 1$ on the test split.

Calibration. Brier score and expected calibration error after isotonic mapping fitted on the calibration split only.

Selective prediction. Risk–coverage curve, its area, and accuracy at coverage $\{0.9, 0.8, 0.5\}$ [Geifman and El-Yaniv, 2017].

Conformal abstention. For each signal, a threshold calibrated on the calibration split to selective risk $\alpha \in \{0.05, 0.10\}$; we report *achieved* risk and *achieved* coverage on the test split.

Uncertainty. Paired bootstrap over items, $B = 10,000$, for every pairwise difference; we report the difference and its 95% interval, never two marginal intervals [Miller, 2024].

Protocol range. Each elicited signal is reported over the three protocols, with a variance decomposition of AUROC into signal, protocol, model and dataset components.

6 Preregistered hypotheses and decision rules

Fixed before data collection and sealed with git tags (prereg-2026-09-01, -b, -c). Deviations are listed in Section 9.

H1 (primary). $s_{\text{TIF}}^{\text{lr}}$ achieves higher AUROC than verbalized scalar confidence. *Retain the triple only if* the paired bootstrap 95% interval excludes zero under all three protocols and in a majority of model $\times$ dataset cells.

H2 (ablation). $s_{\text{TIF}}^{\text{lr}}$ achieves higher AUROC than s_{TF} . If the interval includes zero, the I channel adds no predictive information, whatever H1 shows.

H3 (reference). Semantic entropy achieves higher AUROC than all elicited arms.

H4 (protocol). The between-protocol range of AUROC for a given elicited signal is smaller than the between-signal difference at fixed protocol.

H5 (decision). At guaranteed selective risk $\alpha = 0.10$, the triple yields higher achieved coverage than the scalar.

Reporting commitment. All five hypotheses are reported with effect sizes and intervals regardless of direction. The plan stated that if H1 and H2 both fail, the abstract would say so. Both fail; the abstract says so.

7 Results

Data quality. Parse rates for the elicited arms were at or above 0.90 in 140 of 144 arm $\times$ cell combinations; the four exceptions ($P(\text{True})$ P3 on Claude/ChaosNLI 0.83 and Llama/SciQ-MC 0.88; scalar P2 on Claude/NQ-open 0.88; label-disambiguated triple P3 on Llama/SciQ-MC 0.84) are reported as format-fragile and not imputed. Base error rates were 0.225 (TriviaQA), 0.625 (NQ-open), 0.308 (ChaosNLI) and 0.048 (SciQ-MC; per model 0.037–0.067). SciQ-MC therefore falls below the 5% floor fixed in the plan for three of four models: its AUROC estimates are unstable and we report them with that caveat rather than dropping the regime.

Table 1: Discrimination, pooled over the 16 model $\times$ dataset cells. Mean test-split AUROC for predicting incorrectness, with the range over cells in brackets. Elicited arms are shown per protocol; sampling arms have no protocol. Sequence log-probability is available in 8 cells only.

Arm	Signal	P1	P2	P3
A	$s_{\text{TIF}}^{\text{lr}}$ (fitted)	0.646 [0.39, 0.85]	0.700 [0.54, 0.90]	0.646 [0.52, 0.89]
A	s_{TIF} (fixed weights)	0.653 [0.52, 0.79]	0.697 [0.51, 0.90]	0.642 [0.50, 0.88]
A	s_{TF} (no I)	0.654 [0.52, 0.79]	0.703 [0.51, 0.90]	0.627 [0.49, 0.87]
B	Verbalized scalar confidence	0.657 [0.45, 0.89]	0.710 [0.50, 0.87]	0.658 [0.50, 0.89]
C	$P(\text{True})$	0.614 [0.51, 0.80]	0.603 [0.51, 0.72]	0.652 [0.44, 0.87]
D	Sequence log-probability (8 cells)		0.796 [0.56, 0.93]	
E	Discrete semantic entropy		0.744 [0.56, 0.93]	

Table 2: Mean test-split AUROC by task regime (elicited arms averaged over protocols and models; sampling arms over models). SciQ-MC has a base error rate below the preregistered 5% floor.

Signal	TriviaQA (generation)	NQ-open (generation)	SciQ-MC [†] (clean labels)	ChaosNLI (ambiguous labels)
$s_{\text{TIF}}^{\text{lr}}$	0.711	0.589	0.764	0.592
s_{TF}	0.708	0.598	0.750	0.588
Verbalized scalar	0.707	0.630	0.800	0.564
$P(\text{True})$	0.641	0.606	0.682	0.563
Sequence log-probability	0.869	0.829	0.855	0.632
Semantic entropy	0.825	0.802	0.758	0.591

H1: the triple does not beat one number. Table 3 gives the preregistered contrasts. The mean paired difference in AUROC between the fitted triple and the verbalized scalar is -0.011 , -0.011 and -0.010 under P1, P2 and P3. The 95% interval excludes zero in favour of the triple in 2 of 48 cells and against it in 6; the difference is positive in exactly 50% of cells. The retention rule (interval excluding zero under all three protocols, majority of cells) is not met under any protocol. By regime: TriviaQA $+0.005$, NQ-open -0.041 (four cells significantly against the triple), SciQ-MC -0.033 , ChaosNLI $+0.028$ (one cell significantly in favour). Excluding SciQ-MC, whose base error rate falls below the preregistered floor, the pooled difference is -0.003 over 36 cells. The between-cell standard error of the pooled mean is 0.011, so an average advantage of 0.03 AUROC — a third of what semantic entropy delivers — would have been detected. **H1 fails.**

H2: the indeterminacy channel adds nothing. The fitted triple versus s_{TF} differs by -0.007 , -0.003 and $+0.019$ (P1–P3). Under P1 and P2 no cell favours the triple significantly; under P3 four of sixteen cells do, all on the closed-label datasets. Over all four regimes the mean difference is $+0.003$ (between-cell standard error 0.007). The preregistered rule asks whether the interval includes zero: it does under P1 and P2, and the P3 gain is small and does not transfer to the primary contrast, which fails under P3 as well. **H2 fails.** We conjecture, without testing it here, that the verbal-grade format of P3 spreads I over more of its range (Section 7.1) and that the fitted combination exploits this on three- and four-label tasks; we report it rather than explain it.

H3: sampling bounds elicitation. Semantic entropy exceeds the fitted triple by $+0.096$, $+0.043$ and $+0.097$ (P1–P3), positive in 75% of cells and significantly so in 22 of 48. The advantage is concentrated in free-form generation (TriviaQA $+0.11$, NQ-open $+0.21$, significant in

Table 3: Preregistered contrasts: paired differences in test-split AUROC, mean over the 16 cells per protocol, with the number of cells whose 95% paired-bootstrap interval excludes zero in favour of the first signal (+) or against it (−).

	Contrast	P1	P2	P3
H1	$s_{\text{TIF}}^{\text{lr}} - \text{verbalized scalar}$	−0.011 (+1/ − 4)	−0.011 (+0/ − 1)	−0.010 (+1/ − 1)
H2	$s_{\text{TIF}}^{\text{lr}} - s_{\text{TF}}$	−0.007 (+0/ − 2)	−0.003 (+0/ − 0)	+0.019 (+4/ − 0)
H3	semantic entropy $-s_{\text{TIF}}^{\text{lr}}$	+0.096 (+10/ − 0)	+0.043 (+6/ − 2)	+0.097 (+6/ − 1)

Table 4: Variance decomposition of test-split AUROC (H4): marginal η^2 by design factor, over all signals, protocols, models and datasets. Components are marginal and need not sum to one.

Component	Marginal η^2	Levels
Between datasets	0.281	4
Between signals	0.168	9
Between elicitation protocols	0.024	3
Between models	0.011	4

all 12 NQ-open cells); on the closed-label datasets the two are indistinguishable (SciQ-MC −0.006, ChaosNLI 0.000), where ten samples over three or four labels carry little information. Where available, sequence log-probability is the strongest signal of all (0.80 pooled; 0.83–0.87 on generation).

H4: protocol does not dominate signal, but it moves the numbers. The marginal η^2 of AUROC is 0.28 for dataset, 0.17 for signal, 0.024 for protocol and 0.011 for model (Table 4). The between-protocol component is an order of magnitude smaller than the between-signal component, so the ranking of elicited signals is identifiable and **H4 holds**. Protocol nevertheless shifts the elicited values substantially without changing their ranking: under P3 the share of items with $I \geq 0.5$ rises from 0–4% (P1–P2) to 27–63% on ChaosNLI, and the fitted triple’s AUROC moves by up to 0.10 within a model between protocols. The verbal-grade format makes models *use* the indeterminacy channel without making it informative.

H5: no coverage advantage at guaranteed risk. At $\alpha = 0.10$, in the 36 cells where both signals admit a feasible threshold, achieved coverage of the fitted triple minus that of the scalar averages −0.026 (+0.013 under P2, −0.020 under P1, −0.071 under P3); the triple covers more in 8 cells, less in 23, and the same in 5. **H5 fails**. Two caveats apply to both signals equally: with 150 calibration items the achieved test risk exceeds the nominal 0.10 in about half of the cells (47% for the triple, 56% for the scalar), and semantic entropy gains +0.11 coverage over the scalar on the same cells (18 up, 12 down).

7.1 Post-hoc analyses (registered before collection, not part of the verdict)

Label-semantics collision. In the pilot, applying the arm-A prompt to ChaosNLI produced triples that rated the *meaning of the label* rather than the correctness of the choice: an answer of “contradiction” yielded ($T \approx 0, F \approx 0.9$), “neutral” yielded $I \approx 0.5$ –0.7, and “entailment” yielded $T \approx 1$, item after item. We kept the preregistered prompt and added, on the two closed-label datasets only, a variant that names the object of the rating explicitly (“rate the correctness of the choice, not the meaning of the label”). The disambiguated triple is *worse*: against the scalar,

Table 5: ChaosNLI: Spearman correlation with human label entropy ($n = 300$ per model). I = elicited indeterminacy (arm A, P1/P2); scalar = $1 -$ verbalized confidence (P1); H_c = entropy of ten samples; I_c = constructed residual share.

Model	I (P1)	I (P2)	$1 - \text{scalar}$ (P1)	H_c	I_c
Claude Haiku 4.5	0.31	0.36	0.42	0.18	—
Gemini 2.5 Flash	0.06	0.21	0.20	0.27	0.08
gpt-4o-mini	-0.01	-0.18	-0.01	0.22	0.04
Llama 3.1 8B	0.07	0.08	0.01	0.34	0.30

-0.102, -0.023 and -0.083 (P1-P3); against the original triple, -0.107, -0.016 and -0.066. Telling the model what to rate does not remove the collision and removes what little signal the format had.

Does any indeterminacy track human disagreement? ChaosNLI provides, for each item, the entropy of 100 human labels — a target other than error that an indeterminacy component could plausibly track. Table 5 reports Spearman correlations. The elicited I correlates with human entropy at 0.31–0.36 for Claude (P1–P2), 0.06–0.21 for Gemini, 0.07–0.10 for Llama, and *negatively* for gpt-4o-mini (-0.18 under P2–P3, the label collision again). In every model the scalar confidence tracks human entropy at least as well as I (Claude 0.42). A triple *constructed* from the ten samples (T_c = share supporting the answer, F_c = share of the strongest rival, I_c = residual share) does no better than the scalar entropy of the same samples: the partial correlation of I_c or F_c with human entropy given sample entropy is not distinguishable from zero in any model ($p \geq 0.07$). Finally, the models’ sample distributions diverge from the human distributions precisely on the ambiguous half (Jensen–Shannon 0.26–0.32 versus 0.10–0.13 on clear items; Llama reversed): the models collapse onto one label where annotators split [Baan et al., 2022].

Effective dimensionality. Under P1 and P2 the share of items with $I > T$ is 0–16% in every cell except gpt-4o-mini on ChaosNLI (41–47%, driven by the label collision), and the fitted triple’s AUROC equals that of $F - T$ to the third decimal in most cells. The elicited triple behaves as one degree of freedom dressed as three.

8 Discussion

The plan drafted two discussion branches before the data existed. The preregistered tests select the negative branch; we reproduce it here, and mark separately what rests on post-hoc evidence.

Elicited triples do not predict error better than a single elicited number. This held in free-form generation, in closed-label classification with clean labels, and in closed-label classification with genuinely ambiguous items, on four models from four vendors, under three elicitation formats. The reported prevalence of $T + I + F > 1$ in earlier work [Leyva-Vázquez and Smarandache, 2026, Mason, 2026] therefore describes a property of unconstrained elicitation rather than a usable signal, and any downstream argument that builds on the elicited triple as a measurement instrument lacks predictive support.

The indeterminacy channel adds no predictive information. This is the preregistered result H2: I adds nothing to T and F for predicting error under P1 and P2, and a small protocol-specific gain under P3 (+0.019, four cells) that does not carry to the primary contrast. The

post-hoc ChaosNLI analysis (Section 7.1), which was registered before collection but is not part of the verdict, points the same way: *I* does not track human ambiguity better than a scalar confidence, and a triple constructed from samples is the sample entropy under another name. We state the post-hoc part as consistent with H2, not as established by it.

What elicitation is good for. The verbal arms are not noise: they discriminate at 0.60–0.80 and, in one model, the verbal channel tracks item ambiguity better than the model’s own sampling (0.42 versus 0.18). But the scalar does this as well as the triple. Verbalized reports remain a legitimate object of study as *behaviour* — what a model says under a protocol — and a poor instrument for what it knows, consistent with the observability argument of Mason and Anand [2026].

Where a multi-component representation might still earn its place. Not in self-report. Three routes remain open and are not tested here: components read from internal representations rather than asked for; components assigned by an analyst aggregating several models or sources, where independence of support and conflict is a modelling choice rather than a claim about introspection; and targets that are themselves typed, provided the type labels come from outside the model. The last route is the one ChaosNLI approximates, and on it the elicited *I* did not deliver.

9 Limitations and deviations

$N = 300$ per dataset rather than the planned 1,000, for cost. Per cell (150 test items) the paired-bootstrap intervals on an AUROC difference have a mean half-width of 0.11 (H1) and 0.04 (H2), so single cells are underpowered for small effects; the study’s power comes from pooling 48 cells, where the standard errors of the mean differences are 0.011 (H1) and 0.007 (H2). A true average advantage of 0.03 AUROC for the triple would have been detected; advantages below that are not excluded and would not, in our judgement, justify three numbers over one. Refusal and error rates per cell are given in Appendix B. Short-form and closed-label tasks only; results may not transfer to long-form generation. Discrete semantic entropy over normalized strings is weaker than NLI-based clustering and is uninformative on closed-label tasks. Sequence log-probabilities were available for one model in full and one in part. Three protocols is a small sample from a large space; the protocol range is a lower bound on protocol sensitivity. Elicitation is behavioural: nothing here licenses claims about internal representations. SciQ-MC’s base error rate fell below the preregistered 5% floor.

Deviations from the sealed plan, all recorded before collection: (i) dataset loader rewritten to read parquet files directly; (ii) **gemini-2.0-flash-001** replaced by **gemini-2.5-flash** (no longer served); (iii) items collected in parallel threads; (iv) ChaosNLI added as a fourth dataset with a registered post-hoc analysis of *I* against human entropy; (v) SciQ served as multiple choice after the pilot showed its free-form gold answers (cloze fragments) to be ungradable by exact match; (vi) a label-disambiguated triple added on the closed-label datasets after the pilot exposed the label-semantics collision, reported separately and excluded from the verdict; (vii) N reduced to 300; (viii) SciQ-MC retained although its base error rate (0.037–0.067 by model) fell below the 5% floor at which the plan said a dataset should be dropped — this is the one deviation recorded after collection, because the rate is only known once the greedy answers exist; we keep the regime, flag its AUROC estimates as unstable, and report the pooled contrasts with and without it (H1: -0.010 versus -0.003 ; H2: $+0.003$ versus -0.001 ; H3: $+0.079$ versus $+0.107$), which do not change any verdict. Deviations (iv) and (vi) were expected to favour the triple; (vi) did the opposite. None of (i)–(vii) was made after seeing test-split results.

10 Broader impact

The practical use of any of these signals is to decide when a system should abstain. A signal that looks richer than it is invites over-trust in exactly the settings — medical, legal, financial — where abstention matters most. Our result argues against deploying elicited multi-component states as abstention criteria on the strength of their form, and for the cheaper and better-supported alternatives (sampling, log-probabilities) where they are available. We see no direct risk from the release of prompts and generations; the datasets are public and contain no personal data.

11 Conclusion

Under a preregistered decision rule, on four models and four datasets spanning generation and classification, an elicited three-component epistemic state does not predict error better than a single verbalized confidence number (H1), its indeterminacy component adds no predictive information over truth and falsity (H2), and sampling-based signals dominate all elicited ones on free-form generation (H3). The post-hoc evidence is consistent: as elicited, the triple behaves as one degree of freedom. Claims that rest on the elicited triple as a measurement of what a model knows should be read as claims about what the model says.

Reproducibility statement

Prompts, raw generations (4,800 item-cells), gradings, per-item scores, the sealed analysis plan with its deviation log, and the analysis scripts are provided in the accompanying repository ([\[repository URL\]](#)). The plan and its pre-collection deviations were committed and tagged at 09:03, 09:06 and 09:09 local time on 1 September 2026 (`prereg-2026-09-01`, `-b`, `-c`); the debugging pilot ran between 09:04 and 09:09 and was discarded; full collection ran from 09:10 to 10:29; the analysis outputs were tagged at 10:36 (`results-2026-09-01`). The git history with these timestamps is part of the release, and we acknowledge that a same-day timeline is a weaker guarantee than an external registry; an OSF registration of the plan file is [\[to be added before submission\]](#).

A Elicitation prompts

All prompts are reproduced verbatim from `prompts.py` (version `p2-v1.1`). Placeholders `{question}` and `{answer}` are filled with the item and the model’s greedy answer.

Answer call (all arms). System: *You are answering short factual questions. Reply with the answer only: no preamble, no explanation, no punctuation beyond what the answer needs. If you do not know, reply exactly: I DON’T KNOW.* User: *Question: {question} Answer:*

Arm A, P1. System: *You report epistemic states as three independent numbers.* User: *Question: {question} / Proposed answer: {answer} / Rate three independent degrees, each in [0,1]. They need not sum to 1. T = degree to which the proposed answer is true; I = degree of indeterminacy (vagueness, missing information, undecidability); F = degree to which the proposed answer is false. Reply with exactly one line in this format and nothing else: T=*jnumber* I=*jnumber* F=*jnumber**

Arm A, P2. Same system. User: *Question: {question} / Proposed answer: {answer} / First, in at most three sentences, state what you actually know about this question, what is uncertain, and why. Then, on a final separate line, give three independent degrees in [0,1] which need not sum to 1: T (truth of the proposed answer), I (indeterminacy), F (falsity). The final line must have exactly this format: T=jnumber; I=jnumber; F=jnumber;*

Arm A, P3. System: *You report epistemic states using verbal grades.* User: *Proposed answer: {answer} / Question: {question} / Assign one verbal grade to each of the three dimensions below. The dimensions are independent: a high grade on one does not lower the others. Scale: NONE / VERY LOW / LOW / MEDIUM / HIGH / VERY HIGH / TOTAL. FALSITY – the proposed answer is wrong; INDETERMINACY – the matter is vague, unsettled or unknowable; TRUTH – the proposed answer is right. Reply with exactly three lines and nothing else: FALSITY=jgrade; INDETERMINACY=jgrade; TRUTH=jgrade;* Grid: NONE 0, VERY LOW 0.15, LOW 0.3, MEDIUM 0.5, HIGH 0.7, VERY HIGH 0.85, TOTAL 1.

Arm B, P1. System: *You report confidence as a single number.* User: *Question: {question} / Proposed answer: {answer} / How confident are you that the proposed answer is correct? Reply with a single integer from 0 to 100 and nothing else.*

Arm B, P2. Same system. User: *... First, in at most three sentences, state what you actually know about this question, what is uncertain, and why. Then, on a final separate line, give your confidence that the proposed answer is correct as an integer from 0 to 100, in exactly this format: CONFIDENCE=jinteger;*

Arm B, P3. System: *You report confidence using verbal grades.* User: *Proposed answer: {answer} / Question: {question} / How likely is it that the proposed answer is CORRECT? Scale: NONE / VERY LOW / LOW / MEDIUM / HIGH / VERY HIGH / TOTAL. Reply with exactly one line and nothing else: CORRECTNESS=jgrade;*

Arm C, P1. System: *You verify proposed answers.* User: *Question: {question} / Proposed answer: {answer} / Is the proposed answer true? Reply with a single word, True or False, and nothing else.*

Arm C, P2. Same system. User: *... Check the proposed answer in at most three sentences, then give your verdict on a final separate line in exactly this format: VERDICT=jTrue or False;*

Arm C, P3. Same system. User: *Proposed answer: {answer} / Question: {question} / What is the probability that the proposed answer is TRUE? Reply with a single number between 0 and 1 and nothing else.*

Arm A-label (post hoc, closed-label datasets). As arm A, with “Task:” and “Chosen label:” in place of “Question:” and “Proposed answer:”, and the sentence *The chosen label is one option from a fixed set of classification labels. Rate the CORRECTNESS OF THE CHOICE, not the meaning of the label itself*, with T/I/F redefined as the degree to which the chosen label is the correct classification / indeterminacy about which label is correct / the degree to which the chosen label is the wrong classification.

B Per-cell refusal and error rates

Refusal rate (%) and, among non-refused items, error rate, by model and dataset.

Model	Refusal (%)				Error rate			
	TriviaQA	NQ-open	SciQ-MC	ChaosNLI	TriviaQA	NQ-open	SciQ-MC	ChaosNLI
Claude Haiku 4.5	5.0	18.3	0.3	0.0	0.186	0.645	0.040	0.233
Gemini 2.5 Flash	2.0	13.3	2.0	0.0	0.153	0.531	0.037	0.277
gpt-4o-mini	3.3	18.0	0.7	0.0	0.203	0.610	0.047	0.280
Llama 3.1 8B	6.3	15.7	0.0	0.0	0.363	0.715	0.067	0.443

C Parsing

Numeric triples are parsed by the regular expression $T=<n> \ I=<n> \ F=<n>$ (line breaks tolerated); verbal grades by exact match on the grid; unparseable output is recorded as missing, never imputed.

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